

Communication, Learning, and Retrieval in Transactive Memory Systems

Andrea B. Hollingshead

*Departments of Speech Communication and Psychology,
University of Illinois at Urbana-Champaign*

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A laboratory experiment examined the impact of communication during learning and during recall on collective recall. Dating couples and dyads of strangers worked together to learn and later recall words in six different knowledge categories. Consistent with past work on transactive memory (Wegner, Erber, & Raymond, 1991), dating couples recalled more words than strangers when not able to communicate during learning. This effect reversed when they could communicate during learning: Strangers recalled more words than dating couples. Also, dyads in the same communication condition during learning and during recall remembered more words than dyads that learned and later recalled in different communication conditions. The data support the proposition that communication can have important effects on the manner in which knowledge is learned and recalled in transactive memory systems. © 1998 Academic Press

Consider the following circumstance: You and your significant other just finished a delicious meal at a friend's house, and your friend begins to explain how to prepare one of the dishes. Cooking is not something that you know much about or take care of in your household. Your significant other, on the other hand, is an excellent cook and often cooks at home. How will the two of you learn to prepare the dish at home? There are several methods: (1) You can let your significant other learn it, on the assumption that your significant other is the expert on cooking and will attend to the details. (2) You can attempt to learn it yourself, because your significant other earlier in the day told you of a headache and of an inability to concentrate. (3) You and your significant other can talk about how to

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learn different parts of the recipe, perhaps even interactively quizzing one another after learning each step.

All three methods indicate how knowledge in close interpersonal relationships is handled socially, and all three implicate communication. The former two methods implicate communication prior to learning, that is, communication that implicitly defined expectations about who might learn what. The latter method implicates communication at the point of learning and makes responsibility for collective learning explicit. All three methods are relevant to the analysis of group cognition, group information sampling, and transactive memory, which detail the manner in which groups structure and process information (Hinsz, Tindale, & Vollrath, 1997; Resnick, Levine, & Teasley, 1991; Wegner, 1987, 1995). The objective of the present study was to investigate the impact of communication on the strategies that dyads develop to learn new information and how these strategies, in turn, affect how well dyads recall that information.

A general finding in the groups literature is that an interacting group can remember more information than one individual (e.g., Andersson & Ronnberg, 1995; Basden, Basden, Bryner, & Thomas, 1997; Hinsz, 1990; Weldon & Bellinger, 1997). However, the research indicates that interacting groups recall significantly less than nominal groups composed of noninteracting individuals, indicating evidence of "collective inhibition," i.e., the negative impact of communication on collective retrieval (Weldon & Bellinger, 1997). Andersson and Ronnberg (1995) found that collaborating dyads of friends and dyads of strangers recalled less than nominal dyads. However, they found that collective inhibition was less for friends than for strangers. Basden *et al.* (1997) demonstrated that collective inhibition was due to the disruption of the members' individual retrieval strategies during collaboration, and examined two interventions that improved the recall of collaborative groups relative to nominal groups. In their Experiment 3, when group members were asked to retrieve items from nonoverlapping subsets of categories, and in Experiment 4, when groups were asked to organize their recall by category, i.e., to remember all instances within one category before moving to the next, overall recall did not differ significantly between nominal and collaborative groups. Taken together, the findings of these studies suggest that the impact of communication on collective recall is moderated by the nature of the relationship between group members and by their retrieval strategies.

Though a number of studies have examined retrieval processes in groups, no research has directly investigated the impact of communication at the time of learning on the strategies that groups use to encode and retrieve information. Moreover, previous research on transactive memory demonstrates that the communication processes involved in collective retrieval differ depending on the degree to which people know one another (Hollingshead, 1998a, 1998b). The laboratory experiment reported in this article examined learning and retrieval processes in dyads composed of strangers and dating couples. It controlled communication at

two points in time, during learning and during recall, and assessed the impact of relationship and communication on the strategies that the dyads used to learn and remember new information and on their memory performance.

TRANSACTIVE MEMORY

The theory of transactive memory, as a theory of group cognition, details the relationship between the memory systems of individuals and the communication that occurs between individuals (Wegner, 1987, 1995). Transactive memory refers to the idea that people in continuing interpersonal relationships often develop a specialized division of labor with respect to the encoding, storage, and retrieval of information from different substantive domains (Wegner, 1987). Each member of the relationship becomes a specialist in some areas but not others, and all members come to expect each member to be able to process and access information in specific domains. Such specialization reduces the cognitive load of each individual, while providing the dyad or group access to a larger pool of information across domains. Specialization also reduces the wasted cognitive effort represented by overlapping individual knowledge.

The way the information is distributed in the relationship—which person is responsible for acquiring and encoding information of various sorts—is part of its transactive memory system. If that system is to function effectively, each person needs an accurate representation of it to know who to ask for needed information. The extent to which people in relationships know who has (or is expected to have) various information can affect their communication processes and the quality of their decisions (Hollingshead, 1998a, 1998b; Wegner, Erber, & Raymond, 1991).

It is not just who stores information but how it is stored that matters. Information is not usually recorded as is it encountered; it is transformed or encoded in a variety of ways. How the information is encoded is in part determined by group members' specific roles and responsibilities within their transactive memory system. People in relationships develop differentiated knowledge systems by means of a variety of explicit and implicit procedures. Individuals are seen as linked to knowledge on the basis of their personal expertise, negotiated agreements, or through circumstantial knowledge responsibility, e.g., the person who most recently encountered a domain of information may also incur responsibility for it (Wegner, 1987).

Though transactive memory systems can operate in all relationships, they should be particularly well developed in close personal relationships in which people share responsibilities, engage in conversations about many different topics, and make joint decisions. They should be least developed among strangers. "Strangers" serve as a baseline for understanding how transactive memory systems operate.

Wegner *et al.* (1991) provided evidence of transactive memory systems in dating couples. They demonstrated that people involved in close relationships have an implicit structure for assigning responsibility for learning new informa-

tion based on their shared conception of each partner's expertise. The data indicated that dating couples, compared to strangers, agreed more about the relative expertise of each partner in different knowledge domains. Also, when dating couples were not given an explicit scheme for learning the words, they recalled significantly more words than strangers. Moreover, a reversal occurred when couples were assigned responsibility for specific categories of words: they recalled significantly fewer words than strangers. The implication is that the imposed organizational structure inhibited the performance of dating couples, because it impeded their implicit and efficient system of encoding information developed over the course of their relationship. However, that same organizational structure aided strangers who did not have such a system. Wegner *et al.* (1991) did not examine communication processes in transactive memory; that is, participants did not talk at any time during the memory task.

Hollingshead (1998b) suggested a qualification of the Wegner *et al.* theory of transactive memory, namely that the cooperative storage and retrieval of information relies on the nature of the communication processes between individuals. Dating couples that worked face to face retrieved previously acquired knowledge more effectively than dating couples that worked via a synchronous computer conferencing system that supported only text-based communication and more effectively than dyads of strangers in either communication media condition. One explanation for that finding may be that dating couples encode important information in the nonverbal and paralinguistic communication cues present in their everyday conversations which they use when they are trying to retrieve information together.

Other studies provided indirect evidence of the positive impact of transactive memory systems on group performance. Groups made better decisions when expertise was explicitly assigned to members (Stasser, Stewart, & Wittenbaum, 1995) and when groups accurately recognized the relative expertise of group members (e.g., Henry, 1995; Littlepage, Robison, & Reddington, 1997). Groups also performed better when members were trained together rather than separately (Hollingshead, 1998c; Liang, Moreland, & Argote, 1995; Moreland, Argote, & Krishnan, 1996).

However, no study has directly examined the impact of communication on how information is encoded and subsequently retrieved in transactive memory systems. The present experiment used a similar experimental paradigm and memorization task as Wegner *et al.* (1991). Two communication conditions were investigated: (1) communication during learning, where half of the dyads were allowed to talk and see one another as they *learned* the words and the other half were not, and (2) communication during retrieval, where half of the dyads were allowed to talk and see one another as they *recalled* the words, and the other half were not. This design provided the basis for testing the impact of communication on both learning and recall in transactive memory.

COMMUNICATION AND LEARNING IN TRANSACTIVE MEMORY SYSTEMS

Communication serves many beneficial functions in the encoding and storage of new information in transactive memory systems. It is the medium for transferring information from one individual to another (Hinsz *et al.*, 1997). It is the means by which individuals can learn about the knowledge, expertise, and relevant experiences of other individuals in the system (Hollingshead, 1998a). Individuals who encounter new information can communicate it to the person in the system who needs to know it, is expected to have responsibility for that sort of information, or is most likely to remember it based on previous experiences or personal expertise.

Members of transactive memory systems can also explicitly delegate or negotiate responsibility for learning and storing new information with one or more members through communication (Wegner, 1987). Discussing associations or inferences based on new information can facilitate learning and storage of it by individuals and can help to create a shared representation (Hatano & Inagaki, 1991). At a later time when the dyad or group is trying to access that shared representation, each member's recollections of it may help trigger the retrieval of other aspects of that representation by other members (Hollingshead, 1998a).

However, communication can also hinder the encoding and storage of new information in transactive memory systems. New information may be transferred or explicitly delegated to the "wrong" individual in the system, e.g., one who does not have responsibility for that type of information or is unlikely to remember it due to a lack of expertise, experience, or interest. Discussing associations or inferences based on new information may take time away from learning other important information, especially when individuals must process a large amount of new information or if there is time pressure. Creating a shared representation may not be practical when individuals know that they will need to retrieve information later by themselves without the assistance of other individuals in their transactive memory system (Hollingshead, 1998c; Moreland *et al.*, 1996).

The present experiment investigated the conditions when communication will facilitate and when it will hinder memory performance in transactive memory systems. The memorization task used in this study required dyads to learn a large number of familiar words in a small amount of time. This task was chosen because learning and recall should be best when the dyad has an agreed upon system for dividing and assigning responsibility for different categories of words, i.e., a transactive memory system. It is important to keep in mind that a delegation strategy may not be the most effective for other types of group tasks, for example, when the task is complex and the new information is difficult and unfamiliar.

Memory performance can be conceptualized in several important ways in transactive memory systems. Transactive memory systems develop to reduce the amount of information any one individual needs to know and to provide the group

access to a larger pool of knowledge collectively (Wegner, 1997). The total number of items correctly recalled from the list summed across individuals represents collective knowledge. Total recall can be divided into two categories: unique and overlapping recall. Unique recall refers to those items recalled by only one individual and measures the efficiency of the transactive memory system. Overlapping recall refers to words recalled by both individuals, and measures the redundancy of the transactive memory system. In addition, participants may recall items that are not on the list, which is one measure of the accuracy of information in the system.

Dating couples should recall more words than strangers when they are not able to communicate with their partners at the point of learning because they have a shared conception of relative expertise that will guide how they delegate and assume responsibility for different knowledge categories. This was shown by Wegner *et al.* (1991). Because strangers do not have that shared conceptualization, they will be more likely than dating couples to attempt to learn all words across all categories when they cannot communicate at the point of learning. This will be a less effective strategy that should lead to less unique and more overlapping recall between partners for strangers than for dating couples.

The opportunity to communicate at learning can present a dramatically different condition for both strangers and dating couples. Rulke and Rau (1997) found that newly formed groups spent time discussing their expertise during a training session. Hollingshead (1998b) found that dyads of strangers spent more time discussing their relative expertise than dating couples when working on a knowledge pooling task. Though strangers may make assumptions about what their partner knows on the basis of stereotypes associated with their partner's demographic characteristics such as age, gender, ethnicity, and social class, communication at learning can enable strangers to find out whether those assumptions are accurate, to learn more about one another's expertise, to develop a shared conceptualization of relative expertise, and to explicitly assign responsibility for different categories to each partner. This discussion may lead to explicitly assigning responsibility for different categories to each partner. This should lead to increased total and unique recall and reduced overlapping recall relative to strangers who cannot communicate at the point of learning.

The opportunity to communicate at learning may pose a possible problem for dating couples. If the communication matches the couples' implicit and efficient system for learning new information, then there should be no difference in memory performance for couples who communicate at learning and couples who do not communicate at learning. However, if the couples' communication departs from, or distracts from, their implicit memory system, it may impede memory performance. In other words, communication at learning may disrupt the dating couples' implicit understanding of who knows what, and it may disrupt their implicit procedures for delegating responsibility for different categories of information.

There is reason to believe that communication may impede transactive memory

for dating couples. Wegner *et al.* (1991) found that, for dating couples, the explicit assignment of knowledge categories by the experimenter interfered with learning. Likewise, communication during learning may interfere with the ways that each individual in the relationship implicitly chooses which knowledge categories to learn. Wegner *et al.* (1991) suggested that making an implicit task explicit may have “cognitively disrupted the flow of an otherwise fluid performance . . . like the tennis stroke that falters when it is analyzed” (Wegner *et al.*, 1991, p. 928). Dating couples that communicate may discuss and develop strategies for learning the words in a manner that departs from their implicit system for learning based on differential expertise. These alternative strategies may impede couples’ memory performance. If so, then it follows that couples that communicate at learning, as well as strangers that do not communicate, should also show relative low levels of agreement about the knowledge categories for which each partner assumed responsibility in the task. Couples that do not communicate at learning and strangers that do communicate should show relatively higher levels of agreement. To summarize:

Hypothesis 1. Dating couples will remember more words collectively than strangers when they are not able to communicate as they learn the words. They will show more unique and fewer overlapping words in their individual recall. However, a reversal is predicted when communication is possible: dating couples will remember significantly fewer words than strangers. Therefore, a significant interaction of relationship (dating couples versus strangers) and ability to communicate at learning is predicted for the total recall, unique recall, and overlapping recall.

Hypothesis 2. Dating couples without communication and strangers with communication during learning should agree more about the knowledge categories for which each partner assumed responsibility than dyads in the other conditions. Again, a significant interaction of relationship type (dating couples versus strangers) and ability to communicate at learning is predicted for agreement about who remembered which categories.

Hypothesis 3. The strategies that dating couples and strangers use to remember words will vary depending upon whether they are able to communicate as they learn the words: (a) Dating couples will use delegation learning strategies more often than strangers in the no communication conditions; (b) strangers that communicate at learning will use delegation learning strategies more often than strangers who do not communicate at learning; (c) dating couples that communicate at learning will use strategies other than delegation more often than dating couples that do not communicate at learning.

COMMUNICATION, LEARNING, AND RETRIEVAL

Learning and interacting in groups can facilitate subsequent performance in groups (e.g., Hollingshead, 1998c; Liang *et al.*, 1995; Moreland *et al.*, 1996). Communication can give groups the opportunity to develop a shared representation or shared mental model of their task, e.g., how they will work together to

learn and later remember the words, who will assume responsibility for which categories. Shared mental models provide group members a set of organized expectations for collective performance (Cannon-Bowers, Salas, & Converse, 1993; Oransanu & Salas, 1993). Creating a shared representation may be useful when dyads know that they will have the opportunity to interact with their partner when they need to recall the information. When the dyad is trying to access knowledge based on a shared experience or conversation, each individual's recollections of that event may trigger the retrieval of other aspects of that event by their partner (Hollingshead, 1998a). For example, an individual trying to remember the last few words in a knowledge category may ask her partner for help by referring back to an earlier conversation about that category that occurred during learning. Hence dyads that communicate as they learn the words should recall more words when they can communicate than when they cannot communicate during recall.

However, communication during recall may interfere with performance for dyads who did not communicate during learning. As described earlier, Basden *et al.* (1997) and Andersson and Ronnberg (1995) found that communication during recall tended to disrupt member's individual retrieval strategies. Listening to their partner recall words aloud or responding to their partner's request for assistance may distract individuals from using the strategy they developed at the time of learning (cf. Diehl & Stroebe, 1987). For example, an individual may try to remember the words in a given category in their presentation order just as she did as she was learning them. If her partner interrupts with a question, the individual may forget the other words in the category while responding. Dyads in the no communication during learning conditions should remember more words when they cannot communicate than when they can communicate during retrieval. This reasoning led to the following hypothesis:

Hypothesis 4. Dyads should recall more words collectively when they are in the same communication condition at learning and at retrieval, i.e., communication at both learning and retrieval or no communication at both learning and retrieval. In other words, a significant interaction of the ability to communicate at learning and the ability to communicate at retrieval is predicted for total recall.

METHOD

Overview and Design

Participants learned six sets of six words with a partner. Each set contained one word from each of six knowledge domains: geography, fashion, alcoholic beverages, math, English grammar, and recent movies. The partner was either a significant other or a stranger of the opposite sex from another relationship. After a distractor task, they were asked to recall the words.

The basic design was a $2 \times 2 \times 2$ factorial, manipulating the relationship of the dyad (strangers versus dating), whether they could communicate as they learned the words (yes or no), and whether they could communicate as they recalled the words (yes or no). Participants in the communication conditions were videotaped.

Participants

Eighty-eight heterosexual couples who had been dating for at least 6 months participated in the study. There were approximately 11 couples in each of the eight conditions. Couples were recruited through flyers posted in campus buildings and through presentations made by undergraduate research assistants at weekly meetings of student organizations.

The participants had a mean age of 20.9 years and were either juniors or seniors at the University of Illinois at Urbana–Champaign. Responses to a prequestionnaire indicated that, on average, the couples had been dating 23.6 months and reported seeing each other at least 6 days a week. When asked to rate their relationships on a 1- to 7-point scale, they reported being very committed ($M = 6.7$), very fulfilled ($M = 6.4$), and expected the relationship to last ($M = 6.3$). In addition, they felt knowledgeable about their significant other's areas of expertise ($M = 5.7$) and felt that they communicated well with their significant other ($M = 5.9$). There were no significant differences for these ratings across conditions, reflecting the random assignment of participants to experimental conditions.

Procedure

All participants arrived at the experiment with their significant other. Two to four couples were scheduled for each experimental session. At the beginning of each session, the researcher gave general instructions about the purposes of the study to the participants simultaneously. She explained that the research compared how people in close relationships and people who were previously unacquainted worked together on different types of tasks. She told them that they would work on a word memorization task and would fill out a few questions about their experiences on those tasks. Participants would work either with their significant other, by themselves, or with a stranger on each task. The researcher addressed issues of confidentiality and assured the participants that there would be no deception in the study. She informed them that each person would receive \$5 for their participation and that they could choose to withdraw from the study at any time without penalty. In addition, she explained that a score would be calculated for each participant based on how well each did on the memory task, that the memory task of the pairs who worked together would be combined, and that a bonus of \$20 would be awarded to the pair with the highest score. (The bonus was in fact awarded at the end of the study.)

After the general instructions, participants were randomly assigned to experimental conditions and were led to separate rooms with their assigned partner (either their significant other or a stranger of the opposite sex from another relationship). Strangers were introduced to one another by the experimenter. First, every participant in the study completed a survey that asked about their relationship with their significant other (the person who accompanied them to the study). They were seated in a way to ensure complete privacy. The survey addressed how long they had been dating, how long they expected the relationship to last and how committed they were to it, how well they knew the areas of expertise of their significant other, how well they communicated and made joint decisions, and the activities in which they participated together. Unlike Wegner *et al.* (1991), participants were not asked to rate their own and their partner's relative knowledge in the categories of the word memorization task to avoid the possibility of priming participants to use their relative knowledge as a learning strategy.

The experimenter then gave instructions for the memorization task. First, participants were told that they would work with their assigned partner on the memorization task. The experimenter explained that the objective of the task was to remember as many words as possible from six sets of words from each of six knowledge domains: geography, fashion, alcoholic beverages, math, English grammar, and recent movies. Later in the experiment, participants would be asked to recall as many of the 36 words as possible. Words that each person remembered correctly would be pooled with those that their partner remembered correctly; they would not receive double credit for words that both mentioned. The experimenter also described the communication conditions to which the dyad had been assigned by telling the participants whether they would be able to talk with one another as they learned or as they recalled the words. They were told that their partner received the same instructions.

After the instructions but before the learning phase began, participants were given 1 min to prepare for the memorization task. The minute was given so that participants could think about and discuss (in

the communication at learning conditions) their learning strategy for the task. The researcher did not tell the participants what to discuss during that time. Participants in the no communication at learning conditions sat quietly during the minute.

After the minute ended, the experimenter handed an identical set of six index cards to each partner. Each index card displayed words without listing the knowledge category. Each set of six index cards contained a word or short phrase from each of the six categories: English grammar (e.g., conjunction), recent movies (e.g., *Pulp Fiction*), fashion (e.g., bell bottoms), math (e.g., cosine), geography (e.g., Indonesia), and alcoholic beverages (e.g., margarita). The participants had 1 min to memorize the six words in the set. After each minute, the experimenter collected the set of cards, gave participants a new set, and timed them for another minute. This procedure was followed six times. The word presentation order in the six categories was completely counterbalanced across the six sets of words. A pilot study determined that 1 minute was enough time for collaborating dyads to learn the six words in each set without feeling time pressure.

Participants in the communication at learning conditions were seated at a table at a 45° angle from one another. They were told that they were free to talk during the learning phase. Their interaction was videotaped. Participants in the no communication at learning conditions were seated back to back and away from one another. They were not able to see or to talk to one another during the learning phase. The experimenter emphasized that the dyads would work together on the task, even though they could not communicate. (The video camera was present in both the communication and the no-communication conditions.) The same seating arrangements and instructions were used to manipulate communication at recall.

After participants were presented with the six sets of words during the learning phase, they were given a distractor task to complete individually. The distractor task consisted of logic questions that lasted for 8 min.

During the recall phase, participants were given 4 min to write down all words they could remember from the index cards. Just prior to starting this phase, they were reminded that they were working together, i.e., that their scores for the task would be based on the number of words correctly recalled by either themselves or their partners from the list of 36 words. Afterward, all participants filled out a posttask survey. They responded to one open-ended question about the strategy that they had used to learn the words with their partner and one open-ended question about the strategy that they had used to recall the words with their partner. They also indicated the categories that they had learned and their partner had learned from a list of the six categories. Finally, participants were debriefed fully, thanked, and paid for their participation.

RESULTS

Recall

The measure of dyad recall was based on the total number of words remembered correctly by either or both partners. Unique recall was based on the number of words remembered correctly by only one partner. Overlapping recall was based on the number of words remembered correctly by both partners. Misspelled words were counted as correct, and recalled words not on the list were counted as errors. The highest possible score was 36, and the mean was 27.72. The high score of 35 was obtained by one dyad. The mean confabulation (intrusions) error per dyad was .34 words (about 1%), a very small number, which did not significantly differ across conditions.

A 2 (dating vs. strangers) $\times$ 2 (communication vs. no communication at learning) $\times$ 2 (communication vs. no communication at recall) analysis of variance (ANOVA) was conducted on dyad recall, unique recall, and overlapping recall.

TABLE 1

MEAN TOTAL RECALL AND UNIQUE RECALL AND STANDARD DEVIATIONS AS A FUNCTION OF RELATIONSHIP (DATING VERSUS STRANGERS) AND COMMUNICATION AT LEARNING (NO COMMUNICATION VERSUS COMMUNICATION)

Condition	Total	Unique	Overlapping
Dating, no communication	28.33 _a (2.7)	15.05 _a (9.0)	13.18 _a (10.0)
Dating, communication	27.00 _b (2.9)	15.05 _a (10.5)	11.95 _a (10.5)
Strangers, no communication	27.09 _b (2.9)	11.99 _a (7.9)	16.14 _a (8.7)
Strangers, communication	28.36 _a (3.0)	20.81 _b (7.9)	7.55 _b (6.3)

Note. Dyads are the unit of analysis. Values range from 0 to 36. Dyad *Ns* = 21 to 23. Standard deviations are in parentheses. Means not sharing the same subscript in each column are significantly different, $p < .05$.

In support of Hypothesis 1, dating couples recalled more words than strangers when they could not communicate, and strangers recalled more words than dating couples when they could communicate at learning. The interaction effect of relationship and communication at learning was statistically significant for dyad recall, $F(1, 87) = 3.75$, $p < .05$ (see Table 1). In addition, the interaction effect of relationship and communication during learning was also statistically significant for unique recall, $F(1, 87) = 6.02$, $p < .02$, but was only marginally significant for overlapping recall, $F(1, 87) = 2.75$, $p < .10$. The means indicated that strangers in the communication at learning condition recalled more unique words and fewer overlapping words than dyads in the other conditions.

In support of Hypothesis 4, dyads recalled more words when they were in the same communication conditions during learning and retrieval. The interaction effect of communication at learning and communication at retrieval was statistically significant for dyad recall, $F(1, 87) = 8.30$, $p < .01$ (see Table 2). The interaction effect of communication at learning and communication at retrieval was also statistically significant for unique recall, $F(1, 87) = 15.36$, $p < .001$, and for overlapping recall, $F(1, 87) = 6.97$, $p < .01$. Unique recall was significantly

TABLE 2

MEAN TOTAL RECALL, UNIQUE RECALL, AND OVERLAPPING RECALL AND STANDARD DEVIATIONS AS A FUNCTION COMMUNICATION AT LEARNING (NO COMMUNICATION VERSUS COMMUNICATION) AND COMMUNICATION AT RECALL (NO COMMUNICATION VERSUS COMMUNICATION)

Condition at learning, at recall	Total	Unique	Overlapping
No communication, no communication	28.95 _a (3.0)	16.77 _b (4.5)	12.18 _a (3.3)
No communication, communication	26.36 _b (3.1)	10.32 _c (12.2)	16.05 _a (10.7)
Communication, no communication	27.09 _b (3.1)	14.32 _c (6.0)	12.77 _a (6.4)
Communication, communication	28.27 _a (3.5)	21.54 _a (10.3)	6.72 _b (11.0)

Note. Dyads are the unit of analysis. Values range from 0 to 36. Dyad *Ns* = 21 to 23. Standard deviations are in parentheses. Means not sharing the same subscript in each column are significantly different, $p < .05$.

higher for dyads in the same communication conditions at learning and retrieval than for dyads in different communication conditions at learning and retrieval. Overlapping recall was significantly lower in the communication at learning and communication at retrieval condition than in the other three conditions.

Whether dyads could communicate during learning had a significant impact on both unique and overlapping recall. Unique recall was higher for dyads that communicated during learning than for dyads that did not communicate during learning, $F(1, 87) = 5.57, p < .03; M = 17.93, 13.55$, respectively. Overlapping recall was higher for dyads that did not communicate at learning than for dyads that did communicate, $F(1, 87) = 6.29, p < .02; M = 9.75, 14.11$, respectively. However, the main effect of communication at learning was not statistically significant for dyad recall.

The main effects of partner, and communication at recall, the interaction effect of partner and communication at recall, and the three-way interaction effect were not statistically significant for dyad recall, unique recall, or overlapping recall; all p 's $> .20$.

Category Assignment

Two measures of agreement were generated from a postexperiment survey that asked participants to circle the categories that they had learned and those categories that they believed their partner had learned from the list of six knowledge categories.

The first measure was the number of categories on which both partners agreed that a given partner had assumed responsibility; i.e., both partners circled a given category for the same person. The second measure was the number of categories on which partners disagreed about whether a given partner had assumed responsibility; i.e., only one partner circled the category for a given person. The instances where both people had left the category blank for a given person were not counted as agreements or as disagreements. It was possible that leaving the category blank for one's partner indicated uncertainty about whether the partner had learned the words in that category, as might be the case in the strangers, no communication at learning, no communication at recall condition. A 2 (dating vs. strangers) $\times 2$ (communication vs. no communication at learning) $\times 2$ (communication vs. no communication at recall) ANOVA was conducted on agreement and disagreement separately.

A main effect of communication at learning was obtained for the number of categories on which partners agreed, $F(1, 84) = 10.31, p < .002$. Dyads who communicated during learning agreed significantly more about who had learned which category than dyads who did not communicate during learning ($M = 2.27$ and $.68$, respectively). No other main or interaction effects for this measure were statistically significant.

The results for the number of categories on which partners disagreed were consistent with the findings on dyad recall. A significant interaction effect of relationship by communication at learning was obtained for the number of

categories on which the partners disagreed, $F(1, 84) = 9.18, p < .005$. Strangers in the communication at learning conditions showed less disagreement than strangers who did not communicate at learning, $t(43) = -2.95, p < .005$. However, though the means were in the predicted direction, the amount of disagreement about who learned what did not differ significantly between dating couples in the no communication at learning and communication conditions, $t(43) = -1.44, p > .15$. The four means across the strangers/communication, strangers/no communication, dating couples/no communication, and dating couples/communication conditions were .27, 2.75, 1.80, and 3.04, respectively. This pattern provides some support for Hypothesis 2.

Disagreement was significantly correlated with recall, $r(88) = -.22, p < .04$. The greater the disagreement the worse the memory performance, a result that also supports Hypothesis 2.

Learning Strategies

Self-reported learning strategies collected in the posttask survey were classified into one of three categories: individual, delegation, and other strategies. Individual learning strategies involved learning all words in all categories independently. Delegation learning strategies involved partners dividing responsibility for words in different knowledge categories. Instances of all other attempts at a collective learning strategy were placed into a third category called "other." A sample of each of the three categories of learning strategy, taken from the participants' posttask questionnaires, is shown in Table 3.

The self-reported learning strategies were coded into one of the three categories described above by two independent raters. The coders agreed 97% across the three categories. After all coding was completed, the two raters and the author resolved all discrepancies by consensus.

Table 4 shows the data on the learning strategies. For the statistical analysis of these data, the unit of analysis was the dyad. For each dyad, a count was made of the number of persons in the dyad who reported using each of the three learning strategies; this had a possible range of 0 to 2. These data were subject to a 2 (dating vs. strangers) $\times$ 2 (communication vs. no communication at learning) $\times$ 2 (communication vs. no communication at recall) multivariate analysis of variance, using the individual, delegation, and other learning strategies as the multivariate set. This analysis resulted in statistically significant multivariate main effects for relationship, $F_{\text{mult}}(2, 79) = 6.26, p < .01$, for communication at learning, $F_{\text{mult}}(2, 79) = 68.04, p < .001$, and for communication at recall, $F_{\text{mult}}(2, 79) = 3.12, p < .05$. Also, there was a significant relationship by communication at learning interaction, $F_{\text{mult}}(2, 79) = 6.73, p < .01$. Analysis of the three variates indicated a statistically significant relationship by communication at learning interactions for all three of the learning strategies, for individual, delegation, and other, $F(1, 80) = 4.15, p < .05$, $F(1, 80) = 13.03, p < .01$, $F(1, 80) = 4.91, p < .05$, respectively.

As can be seen in Table 4, the strategies that dyads adopted for learning the

TABLE 3

SAMPLE OF PARTICIPANT STATEMENTS THAT INDICATED INDIVIDUAL, DELEGATION, AND OTHER LEARNING STRATEGIES (AND RELATIONSHIP, COMMUNICATION CONDITION IN WHICH THE STATEMENT OCCURRED)

Individual	1. "I tried to group each word into its category and memorize it. I did not work with my partner at all." (strangers, no communication) 2. "I took the cards and flashed them at a high frequency, low time interval, continuously across my vision and mentally noted which category the word was in." (strangers, no communication) 3. "I didn't work with my partner. I didn't know him either so I tried to remember all the words I could." (strangers, no communication) 4. "I worked alone and went through the cards the whole minute." (strangers, no communication)
Delegation	1. "I memorized my 4 strongest categories: Geography, Math, Movies, and Alcohol. Her stronger categories would be Grammar and Fashion so I barely glanced at those words." (dating couples, no communication). 2. "I tried to think more about words he would have trouble remembering and I tried to memorize those." (dating couples, no communication) 3. "We each were responsible for three groups. I memorized English, movies, and fashion, and she did the other three. This way we were each responsible for less words. (strangers, communication) 4. "Initially, we agreed that he would memorize half and I the other half. The halves were decided based on interests and just splitting them evenly." (strangers, communication)
Other	1. "We tried to come up with stories with people/places we know. Also, after a while he tried to group them into the original categories." (dating couples, communication) 2. "We decided to each look at the word and find which words were important to our lives and remember them in groups of 3. We should have used the method of taking a full category and remembering the words in that category." (dating couples, communication) 3. "We tried to think of ways to put the words together so we could remember them. If we could form sentences that was best, but it didn't always work." (dating couples, communication) 4. "We helped each other with the words we didn't know" (dating couples, communication)

words varied across the experimental conditions in a pattern consistent with Hypothesis 3. Dating couples in the no communication at learning condition were more likely to consider what their partner might know and to delegate categories implicitly than strangers in the no communication at learning condition. The examples from Table 3 demonstrate how some dating couples in the no communication condition focused their efforts on memorizing a subset of categories that were familiar, or thought about the categories that their partners would likely memorize, and selected the remaining categories for themselves. In contrast, strangers in the no communication at learning condition did not consider in their choice of strategy what categories their partners might learn and remember. All strangers in that condition mentioned that they had worked alone or described

TABLE 4

MEAN INDIVIDUAL, DELEGATION, AND OTHER LEARNING STRATEGIES AS A FUNCTION OF RELATIONSHIP (DATING COUPLES OR STRANGERS) AND COMMUNICATION AT LEARNING (NO COMMUNICATION OR COMMUNICATION)

	Dating		Stranger	
	No communication	Communication	No communication	Communication
Individual	1.33 _a	0.18 _b	2.00 _c	0.35 _b
Delegation	0.67 _a	0.68 _a	0.00 _b	1.13 _c
Other	0.00 _a	1.14 _b	0.00 _a	0.52 _c

Note. Dyads are the unit of analysis. Values range from 0 to 2 based on the number of individuals in the dyad reporting having used the strategy, for example, all subjects in the stranger, no communication, condition reported using the individual strategy. Dyad Ns 21 to 23. Means not sharing the same subscript in the same row are significantly different, $p < .05$.

methods to memorize all words in all categories. Examples of individual strategies used by strangers in the no communication condition are in Table 3.

Strangers in the communication at learning condition used delegation more often than dyads in the other conditions, dividing responsibility for the knowledge categories among partners. Dating couples in the communication at learning condition employed other learning strategies more often than in any other condition. Most of these other strategies were collaborative and centered on both members attempting to learn all words together. These strategies were often inefficient, as reflected in the examples in Table 3. Dating couples that used strategies other than delegation often evaluated them negatively or mentioned strategies that they should have used instead.

DISCUSSION

This experiment was the first to directly examine the impact of communication at learning and at retrieval in transactive memory systems. The impact of communication at learning on transactive memory was moderated by the nature of the dyad's relationship and by whether the dyad communicated at retrieval. The findings suggest that communication at learning can facilitate, but can also inhibit the encoding of new information. It also suggests a qualification of the theory of transactive memory proposed by Wegner (1987), namely that the cooperative storage and retrieval of information relies on the nature of the communication processes between individuals. This study strongly supports the statement made by Hinsz *et al.* (1997) that as researchers gain a better understanding of small group communication processes, they gain a better understanding of groups as information processors.

Consistent with Wegner *et al.* (1991), dating couples showed evidence of a transactive memory system when they could not communicate at the time of learning. They remembered more words collectively than strangers. Unlike Wegner *et al.* (1991), in the present study, participants were not asked about their

relative expertise on the knowledge categories before they worked on the memory task, so that they would not be primed to think about relative expertise as a learning strategy. Because the same results were found in this study, it rules out priming as a possible interpretation of the Wegner *et al.* (1991) data.

A reversal of this pattern was obtained when dyads could communicate at learning: dating couples recalled fewer words than strangers. Dating couples in the communication at learning condition tended to develop new strategies for learning the words collaboratively that departed from their implicit knowledge about relative expertise. They also disagreed more about who was responsible for which knowledge categories. It appears that communication about how to learn the words led couples to disregard or to forget about their relative expertise. This is a counterintuitive and important finding, because it suggests that, under some conditions, communication can actually impede the coordination of learning new material in relationships.

Communication facilitated the coordination of learning new information among strangers. When strangers communicated as they learned the words, they were able to delegate responsibility for different knowledge categories explicitly and were able to compensate for the developed transactive memory system of the dating couples. An inspection of the self-reported learning strategies indicated that strangers were much more likely to delegate responsibility for different knowledge categories in the communication at learning condition than in the no communication at learning condition. In fact, no dyads in the strangers, no communication condition mentioned that they considered what their partners knew at all. Each individual attempted to learn all words across all categories in every case. They did not appear to use stereotypes based on their partner's gender or other observable characteristic to infer what their partner might learn and remember and to adjust their learning strategy.

The 1 min of preparation time before the memory task may have helped strangers to develop an effective learning strategy. During this minute, 80% of the stranger dyads in the communication at learning condition discussed and developed a specific learning strategy. In addition, 35% of those stranger dyads explicitly discussed their expertise during that minute: by mentioning their majors, by asking their partners about their expertise in the different categories, or by listing which categories they knew or did not know. In contrast, none of the dating couples in the communication at learning condition made explicit references to relative expertise during that minute. This study demonstrates that when strangers can communicate, they begin to develop a transactive memory system by explicitly establishing relative expertise. This was also found by Hollingshead (1998b). Presumably, as relationships develop, communication provides the basis for learning one another's areas of expertise.

The present study also represents a contribution to the study of group memory. It offers an interpretation of why groups that collaborate at the time of recall remember less than nominal groups. This study showed that dyads develop strategies for organizing and storing information in memory at the time of

learning that are similar to those strategies they think of and use at the time of retrieval. Collective recall was highest for dyads in the same communication conditions at learning and at retrieval. In contrast to previous studies on group memory, there was no significant difference in recall between nominal dyads, i.e., dyads that did not communicate at learning or at retrieval and dyads that communicated at both learning and at retrieval. It appears that when dyads collaborated at both learning and recall, they developed an effective group-level strategy for learning and remembering the words. When dyads were in different communication conditions at learning and at retrieval, their collective recall suffered.

In past studies that examined the impact of collaboration on collective retrieval, the communication conditions at learning and at retrieval were different. Participants were either alone or copresent with other group members and did not collaborate during the presentation of the stimuli. This difference between the communication conditions at learning and at retrieval may offer an explanation for why collaborative inhibition occurred in those studies. It is also interesting to note that no main effects of communication at retrieval and no interaction effects of partner and communication at retrieval were obtained for any of the measures in this study, indicating that conditions at learning had a greater impact on collective memory performance than conditions at retrieval. Clearly, more research is needed to evaluate whether this explanation extends to other tasks and to other situations.

Transactive memory develops in relationships naturally as a way of reducing individual cognitive effort and providing group members with a larger pool of information. In this study, unique and overlapping recall were used to measure efficiency and redundancy. The results showed that communication at learning significantly increased unique recall, reduced overlapping recall, and led to more agreement about who knew what. These findings suggest that communication at learning can increase the efficiency and reduce the redundancy in the transactive memory system and that communication is important for coordinating who will learn what. However, this study also shows that coordinating who will learn what is only one part of an effective transactive memory system; the effect of communication at learning on total recall was moderated by the nature of the dyad's relationship and by whether the dyad communicated at retrieval. Taken together, the results imply that communication at learning will lead to effective memory performance when the discussion about who will learn what corresponds to members' relative expertise. Future research should directly investigate this possibility.

The memorization task used in this research required participants to memorize a large amount of familiar information in a short amount of time. Participants rarely recalled words that were not on the list, perhaps reflecting their familiarity with words in this study. In this specific situation, the most effective strategy for collective recall performance was to delegate responsibility for the knowledge categories between partners. Departures from this strategy, for example, trying to

learn all words oneself as did the strangers in the no communication during learning conditions, or working collaboratively to learn the words together as did some dating couples in the communication during learning conditions, resulted in poorer collective recall.

One should not make the hasty conclusion based on the present study that people in close relationships should not communicate when new information is presented to them. There may be many situations where the collaborative strategies that some dating couples in the communication conditions developed to learn new information would be more successful than delegating responsibility for different knowledge categories to each partner. In the real world, couples are often faced with ambiguous material that they must help one another to encode, and that process probably requires communication. For example, a couple trying to set up their new computer may need to work together to make sense of the instruction manual. In situations where forgetting information can lead to grave consequences, building in more redundancy in the transactive memory system through collaboration at the time of learning may be especially important. Future research should address the effectiveness of individual, delegation, and collaborative learning strategies in other social situations.

This study used self-reports rather than observations of the strategies that dyads used as they learned new information as data. This was done because it was not possible to directly observe the strategies that dyads in the no communication conditions used to learn and remember the words. A comparison of the self-reported learning strategies and the observed learning strategies in videotaped interactions indicated that dyads in the communication conditions, with very few exceptions, accurately represented their learning strategies.

One strength of these experiments is that they directly investigated the impact of other people on individual and group-level cognition in an actual social situation. This work demonstrates the large impact that communicating with others has on the ways that individuals learn, store, and recall information. As such, this research presents a useful, alternative approach for studying the ways that individuals process information gained in social interaction.

The couples in this study were not a representative sample of people involved in dating relationships. Few of the heterosexual couples lived together and none were married. The vast majority of the couples felt very positively about their relationship and expected it to last forever. Future research should examine the generalizability of this study with respect to other types of dating couples, e.g., those who do not feel as positively about their relationships, and other types of relationships, such as coworkers. It is likely that the same transactive effects will be obtained for other types of relationships, which might suggest that intimacy has nothing to do with it—other than possibly motivating people to get to know one another.

Taken together, the results suggest that communication can be a two-edged sword in situations where groups must learn a large amount of information in a small amount of time. It can enhance efficient information processing especially

in new relationships, and it can interfere with expectancies about the division of cognitive labor and produce lower performance in developed relationships. Future research should address whether the results of these experiments are generalizable across such situational factors such as the nature of the new information, the relationship among group members, and the group's processing objective.

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